

**Saraswati Multiple Campus
Lekhnath Marga , Thamel**

BCA / Fourth Semester / CACS 253: Software Engineering

Mid-Term Board Examination

Candidates are required to answer the question in their own words as far as practicable.

Group A (6*5=40)

2. Define Software and Software Engineering. Explain the key challenges of Software Engineering.
3. In the context of Nepal, which software development model is used mostly? Explain with a reason.
4. Write down the differences between cohesion and coupling. Draw a Level 1 DFD for the online shopping system.
5. What is an SRS document, and what are the characteristics of a good SRS document?
6. What are requirements verification and validation?
7. What is a prototype model? List the advantages and disadvantages of the prototype model.
8. "Software does not wear out." Justify the statement.

Group B (2*10=20)

9. Define functional and non-functional requirements. Write down the functional and non-functional requirements of the ATM. Draw a use case diagram.
10. What do you mean by requirements elicitation in the requirements engineering process? What are the methods or techniques used for requirements elicitation? Explain.
11. Discuss different phases of the classical waterfall model. List. What is the lifecycle model? Also, explain what problem would occur if no lifecycle model is followed. List the advantages and disadvantages of the model.

Saraswati Multiple Campus
Lekhnath marga, Thamel
 Mid Term Exam 2080

BCA/Second Year/Fourth Semester

FM: 60 PM: 24

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Group B

Attempt Any Six Questions.

[6 × 5 = 30]

2. What do you mean by true error? Estimate the real root of following non linear equation using bisection method correct upto two significant figures $X^2 \sin x + e^x = 3$.
3. What is interpolation? Given the following set of data points, obtained the table of divided difference. Use the table to estimate the value of $f(1.5)$.

l	0	1	2	3	4
x	1	2	3	4	5
f(x)	0	7	26	63	124

4. Solve the following system of equations by Gauss elimination method.

$$2x + y + 4z = 12$$

$$8x - 3y + 2z = 23$$

$$4x + 11y - z = 33$$

5. Solve the poisson's equation $\nabla^2 f = 2x^2 + y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$.

6. Use the heun's method to estimate $y(0.4)$ when $y'(x) = X^2 + Y^2$ with $y(0) = 0$. Assume $h = 0.2$.

7. Evaluate $\int_0^1 \frac{1}{1-x^2} dx$ by using

i. Trapezoidal rule and ii. Simpson's $\frac{3}{8}$ rule

8. Fit a straight line to the following set of data points:

x	1	3	4	6	8	9
y	1	3	4	4	5	7

Group C

Attempt Any Two Questions.

[2 × 10 = 20]

9. a. Use fixed-point iterative method to approximate the solution of the function $f(x) = \sin^2 x - x^2 + 1$ with $x_0 = 1.5$ and correct up to 5 decimal places.
10. b. Find the Dolittle factorization of the following matrix:

$$\begin{pmatrix} 2 & 5 & 1 \\ 2 & 8 & 0 \\ 8 & 2 & 2 \end{pmatrix}$$

$$5x+2y+2z=18$$

11. Given the data points:

i	0	1	2
x	2	3	4
f	4	9	16

Estimate the function value f at $x=2.5$ using cubic splines.

TRIBHUVAN UNIVERSITY
Faculty of Humanities and Social Sciences
Saraswati Multiple Campus

Time : 3 hrs

Pre Board 2025(Batch 080)

F.M. : 60

Level : BCA \4th Semester

Pass Mark : 24

Code No : CACS302

Subject: DBMS

Attempt any Six questions.

(6 x 5 = 30)

- ~~2.~~ Explain 2NF and 3NF in detail.
3. What is DDL? Explain Types of Constraints with proper example.
[1+4]
- ~~4.~~ What is RDBMS? Explain 3 tier Architecture of DBMS [1+4]
- ~~5.~~ What is Database Model? Explain difference between Network models and Hierarchal Model?[1+4]
- ~~6.~~ What is Weak Entity ? Explain Relational Algebra Operations.[1+4]
7. What is Query Optimization . Expalin in detail.[1+4]
8. What is Join? Explain each term
" INNER JOIN",
" OUTER JOIN",
" Order by "
" GROUP"
"Having" with example. SHOW RESULT IN EXAMPLE.

Group C Long Question

Answer All[1X10=10]

9. Expalin Advantages and Disadvantages of DBMS with example.[5]
 - a. Explain Foreign key, Alternate key, and Candidate key?[5]
10. Explain with example Lost update Problem and Dirty Read with proper example?
 - b)How stored procedure is created explain with example?
11. What is Relational algebra. Explain Select, Project and rename operation in detail with example.

Saraswati Multiple Campus

Mid Term Examination 2082

Candidates are required to answer the question in their own words as far as practicable.

Group B

Attempt all the Questions (6 X 5 = 30)

- 2) Explain process states with its transition.
- 3) Explain dining philosopher problem with semaphore.
- 4) Explain race condition. How can we solve race condition?
- 5) What is file? explain different types of file implementation.
- 6) Differentiate between thread and process. Why do you need threading in multi-program?
- 7) Differentiate between deadlock and starvation. How we can prevent deadlock?

Group C

Answer all the questions. (2 X 10 = 20)

- 8) Schedule the process with following information using FCFS, SJF and RR with quantum time 1.

Process	Arrival Time	Burst Time
P_1	0	7
P_2	2	4
P_3	4	1
P_4	5	4

- 9) Explain Bankers algorithm. Find safe state and unsafe state for the following.

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	0	0	1	2	0	0	1	2	1	5	2	0
P_1	1	0	0	0	1	7	5	0				
P_2	1	3	5	4	2	3	5	6				
P_3	0	6	3	2	0	6	5	2				
P_4	0	0	1	1	0	6	5	6				